DraftBench — Special Education Resource

Curriculum Aligned & Autism Support Framework

Rutherford's Atomic Model

Subject: Physics | **Grade:** 11 | **Duration:** 30 mins

Lesson content

1. Rutherford's Atomic Model

Key idea: Rutherford's model explains atomic structure.

Explanation: Rutherford's experiment showed that atoms have a dense, positively charged nucleus.

• Rutherford discovered the nucleus in the atom.

• Most alpha particles pass through gold foil, but a few bounce back.

Example: Imagine alpha particles like tiny bullets hitting a thin gold sheet.

Key words:

• **alpha particles:** positively charged particles

• **nucleus:** central part of the atom

2. Nucleus

Key idea: Nucleus contains protons and neutrons.

Explanation: Protons and neutrons are found in the nucleus, which is the core of the atom.

• Protons have positive charge.

• Neutrons have no charge.

Example: In a hydrogen atom, there is only one proton in the nucleus.

Key words:

• **protons:** positively charged particles

• **neutrons:** uncharged particles

3. Electrons

Key idea: Electrons orbit the nucleus.

Explanation: Electrons move around the nucleus in orbits, maintaining the atom's stability.

• Electrons have negative charge.

• Electrons are much smaller than protons and neutrons.

Example: In a sodium atom, electrons orbit the nucleus in specific energy levels.

Key words:

• **electrons:** negatively charged particles

• **orbits:** paths around the nucleus

4. Application

Key idea: Understanding Rutherford's model helps explain atomic behavior.

Explanation: Rutherford's model is crucial for understanding atomic interactions and chemical bonding.

• It explains why some alpha particles bounce back.

• It helps predict atomic structure.

Example: Using Rutherford's model, scientists can predict how atoms will combine to form molecules.

Key words:

• **atomic structure:** arrangement of particles in an atom

5. Summary

Key idea: Rutherford's model explains the structure of the atom.

Explanation: Rutherford's model provides a framework for understanding atomic structure and behavior.

• Nucleus contains protons and neutrons.

• Electrons orbit the nucleus.

Example: Understanding Rutherford's model helps in explaining the behavior of elements in chemical reactions.

Key words:

• **protons:** positively charged particles

• **electrons:** negatively charged particles

Teacher support

1. Rutherford's Atomic Model

Teacher Script: Let's start by learning about Rutherford's atomic model. How many of you have heard about Rutherford's experiment before?

Communication Access: Point to a card with the word 'Rutherford' or say the name 'Rutherford' aloud.

Visual Schedule

• **First:** First slide - Rutherford's atomic model (5 mins)

• **Then:** Second slide - Nucleus (10 mins)

• **Last:** Third slide - Electrons (10 mins)

• **Done:** Summary slide - Rutherford's model explains the structure of the atom (5 mins)

IEP Target

English Goal: Some students will correctly provide 80% of the information about Rutherford's atomic model and be able to arrange the atom's structure according to the model.

Nepali Goal: केही बाहरिको वद्यिार्थीले र्फर्ड्र्सको परमिति अणुको मॉडल 80% दरम्यान सही जानकारी दन्छि् र यसको अनुसार अणुको संरचना व्यवस्थति गर्न सक्छन्।

Baseline Probe: In a pre-assessment, 20% of students were able to recall the name of the model but could not explain its components.

Measurable Criteria: 80% accuracy across 5 consecutive sessions

Progress Monitoring: Weekly quizzes and observations during class activities

Accommodations

• Use visual aids and diagrams to support understanding of the atomic structure.

Modifications

• Provide additional time for students to process the information and answer questions.

Prompt Hierarchy

• independent

• verbal

• visual

• gestural

• physical

Reinforcement Plan

Preferred Reinforcer: Access to a favorite educational game

Schedule: Immediate reinforcement after correct responses

First–Then: First answer the question correctly, then you can play your favorite game.

Sensory Accommodations

• Use a calm and quiet environment to reduce sensory overload.

Calm-Down Strategy

Use deep breathing exercises when the student becomes overwhelmed.